REPORT 5 - NOSE CONE 720 MM LOADS FOR STRUCTURE

Fabio Feig - 315169



EPFL ROCKET TEAM

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Weisshorn – 720 mm Nose Cone: Pressure, Forces and Heating

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1 Results for the structure team

This report gives ST the aerodynamic and thermal inputs to size the two 720 mm noses. ST computes the joint and shell loads from them. Both noses are 720 mm long with $D = 180$ mm and a 4.5 mm tip radius; their geometry is in `2027_C_FD_NOSE-CONE-720MM_GEOMETRY`.

- **A** (baseline, 3U inside the nose): LV-Haack, 679.0 mm + 41 mm coupler.
- **B** (backup, 3U elsewhere): Von Kármán, 700.0 mm + 20 mm coupler.

The numbers come from eight Fluent runs (2D axisymmetric RANS, zero incidence) at $M = 0.95$, 1.20 and 1.47 (burnout), and from the Python flight model, calibrated on these runs, for the time histories and the hot day (30 °C, LVO masses). The nose is **sealed**: the air inside stays at the pad pressure $p_{pad} = 101.3$ kPa, raised by its heating,

$$p_{in} = p_{pad}\,\frac{T_{in}}{T_{pad}}, \qquad \Delta p = p_w - p_{in}, \tag{1}$$

between $T_{in} = T_{pad}$ and T_{in} = mean temperature of the inner face of the shell (upper bound). $\Delta p > 0$ pushes the shell inwards. Axial forces are positive when they push the nose aft onto the joint (compression).

Table 1: Load cases. Pressures: tip with p_{in} at T_{pad} (largest inward load), shell minimum with heated air (largest outward load). Net pressure force on the joint: $\sum (p_w - p_{in})\, 2\pi r\, \mathrm{d}r$ over the outer surface, for $T_{in} = T_{pad}$ / heated air. D_n: aerodynamic axial force of the nose (model, 20 µm roughness). N/α: normal force per degree of incidence.

	A nominal	A hot day	B nominal	B hot day
LC1, burnout (M_{max}, q_{max})				
Time t [s], Mach M	9.1, 1.47	9.1, 1.56	9.1, 1.47	9.1, 1.56
Dynamic pressure q [kPa]	116.8	131.4	117.5	132.4
Δp at the tip [kPa]	153	175	154	177
Lowest Δp on the shell [kPa]	-46	-49	-48	-50
Net pressure force on the joint [N]	-397 / -753	-374 / -767	-436 / -793	-418 / -813
Aerodynamic axial force D_n [N]	278	310	234	260
Normal force N/α [N/°]	104	117	104	118
LC2, transonic ($M = 0.95$ on the ascent)				
Time t [s], q [kPa]	5.9, 57.6	5.8, 57.9	5.9, 57.3	5.8, 58.0
Δp at the tip [kPa]	63	64	62	64
Lowest Δp on the shell [kPa]	-22	-21	-29	-28
Net pressure force on the joint [N]	-265 / -358	-248 / -335	-263 / -355	-246 / -333
Aerodynamic axial force D_n [N]	53	51	44	47
Normal force N/α [N/°]	51	51	51	52
LC3, apogee ($q \approx 0$)				
Time t [s], p_∞ [kPa]	43.7, 31.3	44.2, 31.1	43.9, 30.8	44.5, 30.5
Δp on the whole shell [kPa]	-70	-70	-71	-71
Net pressure force on the joint [N]	-1783	-1788	-1794	-1802
Nose centre of pressure x_{CP} from the tip [mm]	295		347	

Table 2: Heating. Fluent values: burnout, nominal, wall at 300 K. Maximum temperatures over the flight: Python model calibrated on Fluent (Section **??**), CFRP shell 1.2 mm, Al 6082 tip insert 80 mm.

	A nominal	A hot day	B nominal	B hot day	Limit
Heat flux at the tip q_w [kW/m^2] (Fluent)	172.5	–	173.4	–	
h at the tip / shoulder [W/(m^{2}K)] (Fluent)	1888 / 546	–	1899 / 539	–	
Stagnation temperature T_0 [°C]	118	154	119	156	
Al tip insert [°C]	61	84	56	79	
Shell outer face, 50 mm [°C]	82	110	82	110	120 (DP490)
Shell inner face, 50 mm [°C]	77	104	78	104	67 (Req_10)
Inside air, upper bound [°C]	72	98	73	99	67 (Req_10)

What this means for the structure.

1. **The sealed nose is the largest load.** At apogee the outside pressure has dropped to 31.3 kPa while the inside stays at the pad pressure: $\Delta p \approx -70$ kPa on the whole shell, which pulls the nose forward off the joint with about 1.8 kN, close to the 2000 N of Req_07. At burnout the inside overpressure already outweighs the drag: the net pressure force is -397 to -753 N (tension). Vent holes would bring p_{in} to p_∞ and remove this load.
2. **Inward pressure** is highest at the tip at burnout, 153 kPa (175 kPa hot day), on the spherical cap only. Beyond $x = 2$ mm it is at most 36 kPa for A and 17 kPa for B (43 and 22 kPa hot day), and below 10 kPa after 50 mm (Figure **??**).
3. **Aerodynamic axial force** peaks at burnout: 278 N for A and 234 N for B (310 and 260 N hot day). Fluent with a smooth wall gives 282.6 and 237.6 N.
4. **Normal force** under incidence α: $N = 2\,q\,A_{ref}\,\alpha$, acting at x_{CP}, distributed as $\mathrm{d}N/\mathrm{d}x = 2\,q\,\alpha\,\mathrm{d}S/\mathrm{d}x$ (Section **??**). ST chooses the design incidence (Req_12).
5. **Temperatures.** The DP490 limit (120 °C) is not reached. The inner face of the shell and the inside air exceed Req_10 (67 °C) already on a nominal day, for both designs.
6. **A against B.** The heating is the same within 1 %. B has 15 % less drag at burnout, about half the inward pressure on the front of the shell (item 2), and a 20 mm coupler instead of 41 mm. The sealed-nose loads are the same for both.

The data behind every figure are delivered as CSV tables (Appendix **??**).

2 Designs and flight points

Table 3: The two 720 mm noses. L: Haack length to the sharp apex; the 4.5 mm spherical tip removes δ, so the curved part is $L - \delta$ long. Apogee: Python flight model calibrated on Fluent (Section **??**).

	A (baseline)	B (backup)
Profile	LV-Haack ($C = 1/3$)	Von Kármán ($C = 0$)
3U payload	inside the nose	not in the nose
L / δ [mm]	682.3 / 3.3	705.5 / 5.5
Curved part x_S + coupler c [mm]	679.0 + 41	700.0 + 20
Fineness L/D	3.79	3.92
Apogee, nominal / hot day [m]	8887 / 9397	8982 / 9518

Flight. The trajectory is the one of the Python design tool (report 2), with the nose of each design. Two cases are given everywhere:

- **nominal**: ISA day (288.15 K on the pad), propellant mass from the total impulse;
- **hot day**: 303.15 K on the pad and the LVO masses. The rocket is lighter and faster, so q and the heating are higher.

Along the trajectory,

$$q = \tfrac{1}{2}\,\rho_\infty V^2 = \tfrac{1}{2}\,\gamma\, p_\infty M^2, \qquad T_0 = T_\infty \left(1 + \tfrac{\gamma-1}{2}\,M^2\right). \tag{2}$$

The highest Mach number, dynamic pressure and stagnation temperature all occur at burnout (Figure 1).

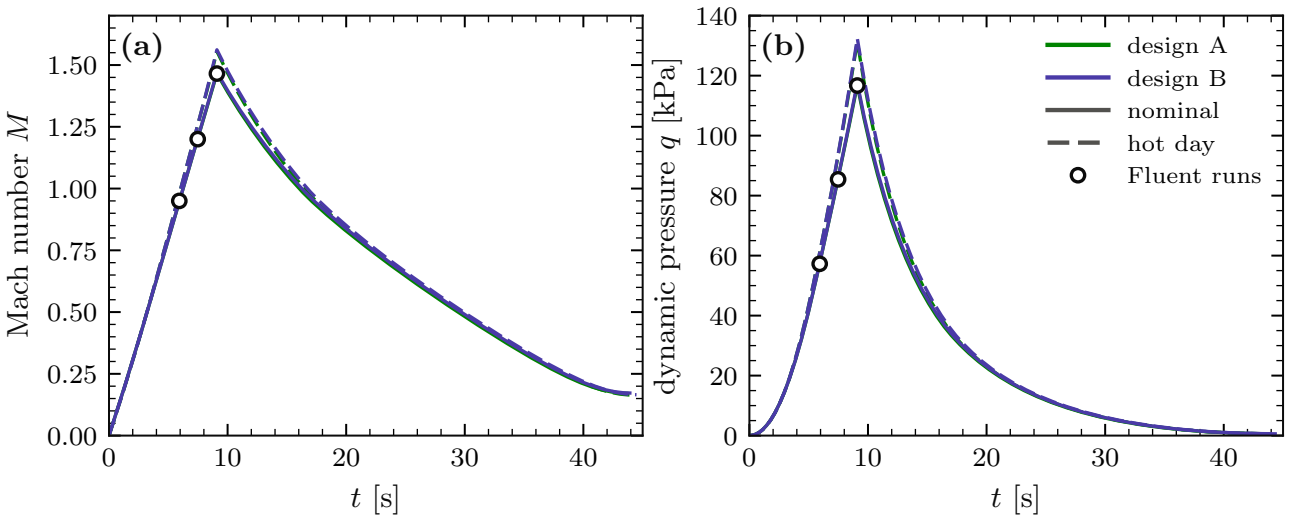


Figure 1: (a) Mach number and (b) dynamic pressure along the flight, both designs, nominal and hot day. Circles: the free stream of the Fluent runs.

Fluent points. The three Fluent runs use the free stream of design A's nominal trajectory at these instants (Table 4); design B is run at the same points, so that A and B differ by shape only (B's own burnout is at $M = 1.471$). Values at other times are interpolated in M (Section **??**).

Table 4: Free stream of the Fluent runs. Reynolds number per metre: $\rho_\infty V/\mu_\infty$.

	Transonic	Supersonic	Burnout
Time t [s]	5.93	7.48	9.10
Altitude z [m]	923	1476	2185
Mach number M	0.950	1.200	1.465
Static pressure p_∞ [Pa]	90714	84801	77685
Static temperature T_∞ [K]	282.15	278.55	273.95
Dynamic pressure q [kPa]	57.3	85.4	116.7
Reynolds number per metre [10^6/m]	20.3	24.4	27.9

Load cases. Three instants bound the aerodynamic loads of the sealed nose:

- **LC1, burnout**: highest M, q, inward pressure at the tip, axial force, normal force per degree and heating;
- **LC2, transonic**, $M = 0.95$ on the ascent: the shock system changes, lower q;
- **LC3, apogee**: $q \approx 0$ and the lowest outside pressure, so the largest outward (burst) pressure on the sealed nose.

Recovery (Req_07) and the inertial loads at the joint are not aerodynamic and are left to ST.